

The cone — surface area and volume

A pyramid whose base is a circle: the same two formulas, with π and a slant height.

LESSON 36–37

2 × 40 MIN

GEOMETRY 11, §17

IGCSE E13.X

LESSON CLOCK

14'

24'

24'

22'

20'

6'

The elements	14 min
The net and the curved area	24 min
The volume	24 min
Working backwards	22 min
Practice	20 min
Homework	6 min

LEARNING OBJECTIVES

- Name the base, axis, height and slant height of a cone.
- Derive the curved surface area from the net.
- Compute the volume of a cone.
- Solve for a missing dimension.

TEXTBOOK

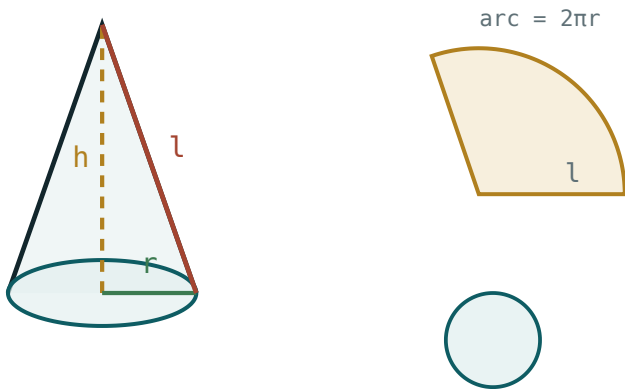
National	pp. 179–190
Cambridge	Core & Extended, pp. 335–342
Workbook	Exercise 17.1

01

Explanation

The elements

A **right circular cone** is swept out when a right triangle turns about one of its legs. Its **axis** joins the apex to the centre of the base circle and is perpendicular to it.



$S = \pi r^2 + \pi r l$ – a circle and a sector

The height, the radius and the slant height form a right triangle. Cut down the slant and it unrolls into a sector.

$$l^2 = h^2 + r^2$$

The **axial section** — the cut through the axis — is an isosceles triangle of base $2r$ and height h , with equal sides l . Every cone question can be reduced to that triangle.

The net and the curved area

Cut the cone down one slant line and unroll it. The curved surface becomes a **sector** of radius l whose arc is the base circumference $2\pi r$. The sector is the fraction $\frac{2\pi r}{2\pi l}$

$= \frac{r}{l}$ of the full circle of radius l :

$$S_{\text{curved}} = \frac{r}{l} \times \pi l^2 = \pi r l$$

$$S_{\text{total}} = \pi r^2 + \pi r l = \pi r(r + l)$$

l , NOT h

The curved surface uses the **slant** height. Using h understates it, and the two are equal only for a flat disc. Compute $l = \sqrt{h^2 + r^2}$ first, every time.

The volume

$$V = \frac{1}{3} \pi r^2 h$$

A cone is exactly one third of the cylinder on the same base with the same height — the same $\frac{1}{3}$ as a pyramid against its prism. Both come from the same integral, $\int x^2 dx$, as Quarter III algebra showed.

SOLID	VOLUME	CURVED / LATERAL AREA
cylinder	$\pi r^2 h$	$2\pi r h$
cone	$\frac{1}{3}\pi r^2 h$	$\pi r l$
prism	Bh	Ph
pyramid	$\frac{1}{3}Bh$	$\frac{1}{2}Pl$

Working backwards

$$h = \frac{3V}{\pi r^2} \quad r = \sqrt{\frac{3V}{\pi h}} \quad l = \frac{S_{\text{curved}}}{\pi r}$$

Example. A cone of volume $100\pi \text{ cm}^3$ and radius 5 cm has $h = \frac{300\pi}{25\pi} = 12 \text{ cm}$, and then $l = 13$ and $S_{\text{curved}} = 65\pi \approx 204 \text{ cm}^2$.

Given the **total** area, the equation is quadratic in r — expand $\pi r(r + l)$ and solve, rejecting the negative root.

02 Worked examples

EXAMPLE 1 A cone has radius 6 cm and height 8 cm. Find its slant height, total surface area and volume.

$$① \quad l = \sqrt{64 + 36} = 10$$

$$② \quad S = \pi(6)(6 + 10) = 96\pi \approx 301.6$$

$$③ \quad V = \frac{1}{3}\pi(36)(8) = 96\pi \approx 301.6$$

A coincidence of these numbers.

ANSWER

$$l = 10; S = 96\pi; V = 96\pi$$

EXAMPLE 2 A cone has volume $100\pi \text{ cm}^3$ and radius 5 cm. Find its height and curved surface area.

$$1 \quad 100\pi = \frac{1}{3}\pi(25)h$$

$$2 \quad h = 12$$

$$3 \quad l = 13$$

$$4 \quad S_{\text{curved}} = \pi(5)(13) = 65\pi \approx 204$$

ANSWER

$$h = 12; 65\pi \approx 204 \text{ cm}^2$$

EXAMPLE 3 A sector of radius 12 and angle 120° is rolled into a cone. Find its radius and height.

$$1 \quad l = 12; \text{arc} = \frac{120}{360} \times 24\pi = 8\pi.$$

$$2 \quad 2\pi r = 8\pi \Rightarrow r = 4$$

$$3 \quad h = \sqrt{144 - 16} = \sqrt{128} \approx 11.3$$

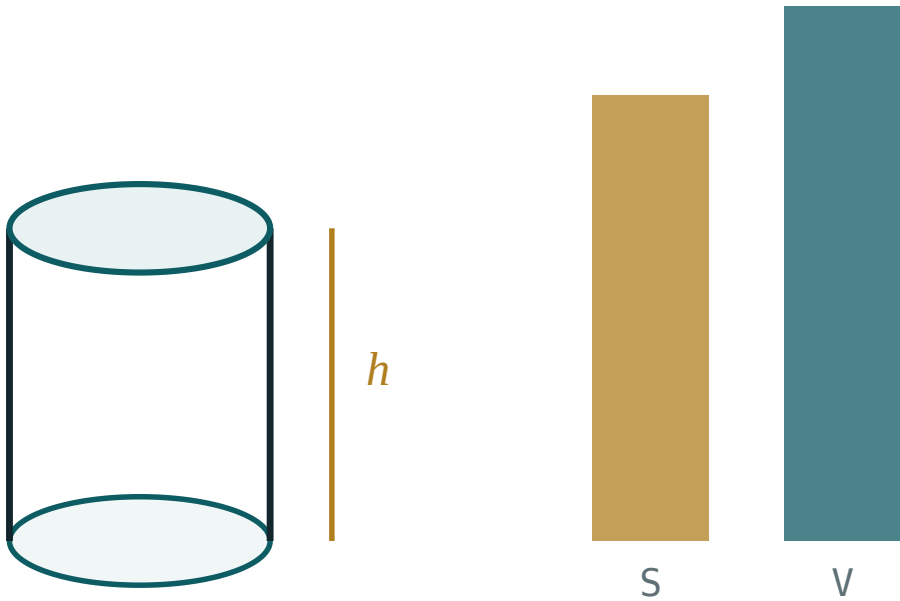
ANSWER

$$r = 4, h \approx 11.3$$

03 Interactive model

Roll a paper sector into a cone and measure the base radius against the arc length.

The cone against the cylinder



base area B **50.3**

perimeter P **25.1**

total S **251.3**

volume V **301.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Cone	Konus	Конус
Apex	Uchi	Вершина
Axis of a cone	Konus o'qi	Ось конуса
Slant height	Yasovchi	Образующая
Curved surface area	Yon sirt yuzasi	Площадь боковой поверхности
Net of a cone	Konus yoymasi	Развёртка конуса
Axial section	O'qli kesim	Осевое сечение
Right circular cone	To'g'ri doiraviy konus	Прямой круговой конус
Solid of revolution	Aylanish jismi	Тело вращения

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

Quick check
Check yourself

The curved surface area of a cone is:

$$\pi r^2$$

$$\pi r l$$

$$\pi r h$$

$$2\pi r l$$

The volume of a cone is:

$$\pi r^2 h$$

$$\frac{1}{3} \pi r^2 h$$

$$\frac{1}{2} \pi r^2 h$$

$$\pi r l$$

l equals:

$$h + r$$

$$\sqrt{h^2 + r^2}$$

$$\sqrt{h^2 - r^2}$$

$$hr$$

The net of a cone is:

a rectangle and two circles

a sector and a circle

two sectors

a triangle

The axial section of a cone is:

a circle

an isosceles triangle

a rectangle

a trapezium

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. l with $r = 3, h = 4$

ANSWER 5

2. l with $r = 5, h = 12$

ANSWER 13

3. Volume with $r = 3, h = 4$

ANSWER 12π

4. Curved area with $r = 3, l = 5$

ANSWER 15π

5. Total area with $r = 3, l = 5$

ANSWER 24π

6. Volume with $r = 6, h = 10$

ANSWER 120π

7. A cone is what fraction of its cylinder?

ANSWER $\frac{1}{3}$

Medium · the standard question

7 PROBLEMS

1. $r = 6, h = 8$: total area and volume

ANSWER 96π each

2. Volume 100π , radius 5: the height

ANSWER 12

3. Same: the curved area

ANSWER 65π

4. Sector radius 12, angle 120° rolled up: r

ANSWER 4

5. Same: the height

ANSWER $\sqrt{128} \approx 11.3$

6. A cone of $r = 7, l = 25$: the height

ANSWER 24

7. A cone of curved area 60π and $r = 5$: l

ANSWER 12

Hard · stretch and reason

7 PROBLEMS

1. A cone of total area 90π and slant height 12: the radius

ANSWER $r = 5$

2. A cone of volume 96π and slant height 10: r and h

ANSWER $r = 6, h = 8$

3. The sector angle for a cone with $r = 4, l = 12$

ANSWER 120°

4. A cone whose axial section is equilateral, $r = 5$: its volume

ANSWER $\frac{125\pi\sqrt{3}}{3} \approx 226.7$

5. A cone of maximum volume from a sector of radius 12: the angle

ANSWER $\approx 294^\circ$

6. Two similar cones with volumes 8 and 27: the area ratio

ANSWER $4 : 9$

7. A cone is inscribed in a cube of edge 12: its volume

ANSWER 144π

07 Homework

Homework — 5 tasks

Compute l before any surface-area formula.

1. A cone has radius 9 cm and height 12 cm. Find its slant height, total surface area and volume.
2. A cone has volume 147π cm³ and radius 7 cm. Find its height and curved surface area.
3. A sector of radius 15 and angle 144° is rolled into a cone. Find its radius, height and volume.
4. A cone has total surface area 144π and slant height 15. Find its radius.

5. Explain in three sentences why the curved surface area uses l and not h .